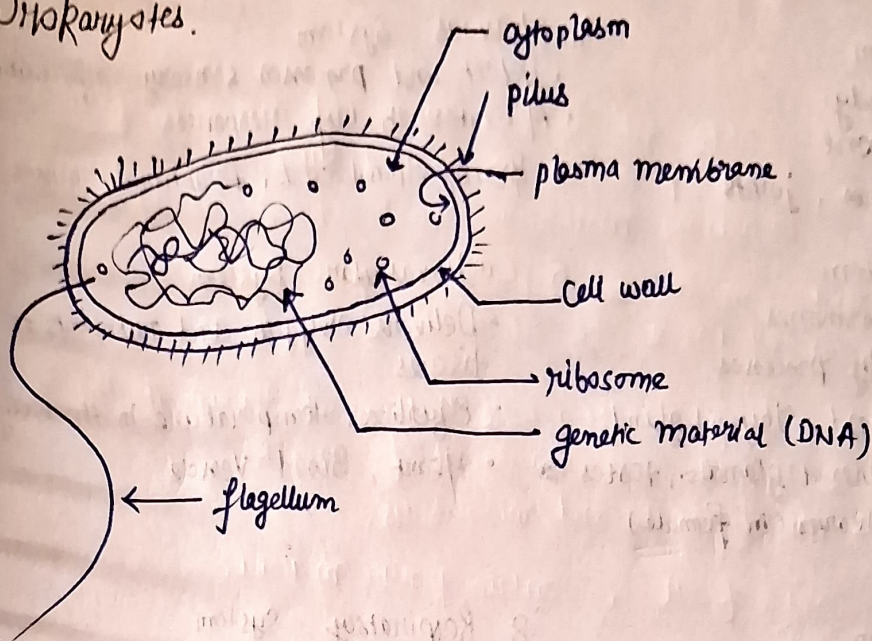
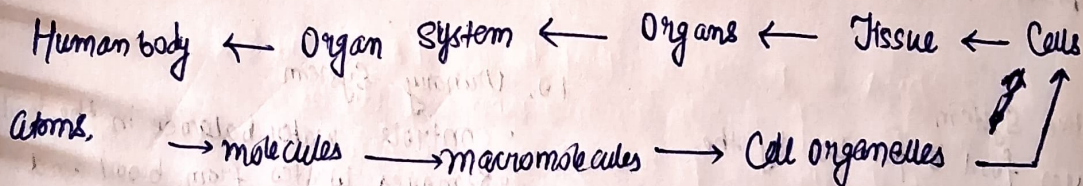


Prokaryotes.



Homo Sapiens (Human)

- 1) Human body is composed of many different types of organ systems
 → They ensure homeostasis and the viability of the human body.



Various body Systems:

1. Respiratory System
2. Cardiovascular System
3. Digestive System
4. Nervous System
5. Lymphatic System (Immune System)
6. Endocrine System
7. Integumentary System (Exocrine System)
8. Musculoskeletal System
9. Urinary System (Excretory system)
10. Reproductive System.

1. Integumentary System
 - Encloses internal body structure
 - Site of many sensory receptors
 - Hair, Skin, Nails.
2. Muscular System
 - Enables movement
 - Helps maintain body temperature
 - Skeletal muscles, tendons.
3. Skeletal System
 - Supports the body
 - Enables movement
 - Cartilages, bones, joints
4. Nervous System
 - Detect and processes sensory information
 - Activates bodily responses
 - Brain, Spinal cord, Peripheral nerves.
5. Endocrine System
 - ~~or~~ secretes hormones
 - Regulates bodily processes.
 - Pituitary gland, Thyroid gland, Pancreas, Adrenal glands, testes (in males), Ovaries (in female)
6. Cardiovascular System
 - Delivers oxygen and nutrients to tissues
 - Equalizes temperature in the body
 - Heart, Blood vessels.
7. Lymphatic System
 - Returns fluid to blood
 - Defends against pathogens
 - Thymus, Lymph nodes, Spleen, Lymphatic vessels.
8. Respiratory System
 - Remove CO_2 and delivers O_2 to blood
 - Nasal passage
 - Trachea
 - Lungs
9. Digestive System
 - Processes food for use by the body.
 - Removes wastes from undigested food.
 - Stomach, Liver, Gall bladder, Large intestine, Small intestine.
10. Urinary System
 - Controls water balance in the body
 - Removes wastes from blood and excretes them
 - Kidneys, Urinary bladder
11. Reproductive System
 - Male

 - Produces sex hormones and gametes
 - Delivers gametes to female
 - Epididymis, Testes.

Female

 - Produces sex hormones and gametes
 - Supports embryo/fetus until birth
 - Produces milk for infant.
 - Mammary glands, Ovaries, Uterus.

Organs

Total = 79 organs

~~Longest~~ longest and longest organ = skin (but outside) } Regenerative
Longest organ inside body = Liver

Nerve cell / Brain cell } Regenerative
Heart cell / Cardiac cell }

Smallest organ - Pineal gland (present near the centre of the brain, in a groove between the hemispheres)

Heart - first major functional organ to develop and starts to beat and pump blood at about 3 weeks into embryogenesis.

Lungs - last major organ to develop and mature before birth.

~~Brain~~ Brain - last to reach its final mature form and develop even after birth till 25 yrs.

Sensory Organs.

Eyes, Ears, Nose, Tongue, Skin.

Mesentery: New organ (78 + mesentery = 79)

- Large single continuous stretch of tissue that supports and position all the digestive organs in the abdomen.
- The organ is responsible for transporting blood and lymphatic fluid between the intestine and rest of the body.

Tissues:

Group of cells with a similar structure working together. Working together for a specific function. There are four main types of tissues in a body:

1. Epithelial tissue - layers of cells that cover the surface of the body that come into contact with the exterior world, lines internal cavities and passageways and form glands
2. Connective tissue - binds the ~~cell~~ cells and organs of the body together and performs many many functions, especially in the protection, support, and integration of the body.
3. Muscle tissue - is excitable, responds to stimulation and ~~contracts~~ contracts to provide movement.
- occur as three major types: skeletal muscles, smooth muscles, cardiac ~~muscles~~ muscles in the heart.
4. ~~Nervous~~ Nervous tissues: is also excitable, allow the body to receive signals and transmit information as electric impulses from one region of body to another.

Cells:

87.2 trillion cells. } excluding microbes living in body
200-220 cell types.
eg- neurons, cardiomyocyte, β -cell, α -cell, RBC, B-lymphocyte, T-lymphocyte.

Cell Organelles:

1. Cell membrane - double phospholipid layer and monolayer of protein scattered around phospholipid layer. The other components in plasma membrane are cholesterol and glycoproteins.
2. Cytosol - fluid that contains ~~cell~~ organelles; represents the material outside the nucleus and inside the cell membrane.
3. Cytoskeleton - network of protein structure that extend throughout the cytoplasm provide the cell with an internal framework.
eg - microfilaments and microtubules.
4. Nucleus - control centre of cell, contain genetic material, three main regions:
 - (i) nuclear membrane
 - (ii) the nucleolus
 - (iii) chromatin
 - Nuclear membrane - barrier of nucleus. It consist of double phospholipid membrane and contains nuclear pores for exchange of material with the rest of cell.
 - Chromatin - DNA + protein scattered throughout the nucleus.
→ condenses to form chromosomes when cell divides
 - Nucleolus - site of ribosome formation, ribosome then migrates through nuclear pores.
5. Ribosomes - site of protein synthesis in the cell.
6. RER - carry ribosomes.
7. SER - cholesterol synthesis and breakdown, fat metabolism and detoxification of drugs.
8. Golgi Apparatus - modifies and packages proteins, secrete vesicles, plasma membrane components and lysosomes.
9. Mitochondria - generate ATP for cellular activities.
10. Lysosomes - suicidal bag of the cell.
contain enzymes that digest non-usable materials of cell

PH - 4.5

11. Peroxisomes: detoxify harmful substances and break down free radicals
Involved in catabolism of very long chain fatty acids.
12. Centrioles: involved in the organization of mitotic spindle fibers during cell division and in the completion of cytokinesis.
13. Centrosomes: serves as the main microtubule organizing centre (MTOC) of animal cells as well as a regulator of cell-cycle progression; are composed of two centrioles. ~~are composed of two~~ arranged at right-angles to each other.
14. Vesicles - involve in metabolism, transport
buoyancy control, temporary storage of food and enzymes.

Macromolecules : Nucleic Acid Lipid
 Carbohydrates Protein.

Number of molecules: 2×10^{25} molecules
and more than 99% of them are water

Number of atoms: 7×10^{27} atoms. for 70kg Adult.

~~the~~ No. of atoms in Human Cell = 10^{14}

C, H, N, O, Ca, P } 99% of atoms in a person

Note: RBC & Platelets } no DNA

mitochondria / Chloroplast - their own DNA

RBC - biconcave shape $\Rightarrow$ increases Surface Area to Volume ratio of RBCs
due to large surface Area it facilitates rapid diffusion of oxygen and CO_2 across the RBC's plasma membrane.

How many Haemoglobins are in 1 RBC? 270 million.

why RBC live for 120 days only? RBCs accumulate wear and tear over time
they become less efficient at job
Macrophages then digest RBCs

Prokaryotes

Pro + Karyon
Primitive nucleus

- The presence of
- i) Absence of well-defined nucleus
- ii) Absence of membrane-bounded organelles
- iii) Absence of cyto-skeleton
- iv) Unicellular
- v) Circular DNA
- Single chromosome.
- vi) Smaller in size (1-10 μ m)
- vii) base pairs: $10^6 - 5 \times 10^6$
- viii) reproduce asexually only
- ix) Presence of plasmid

What is cytoskeleton? Examples.

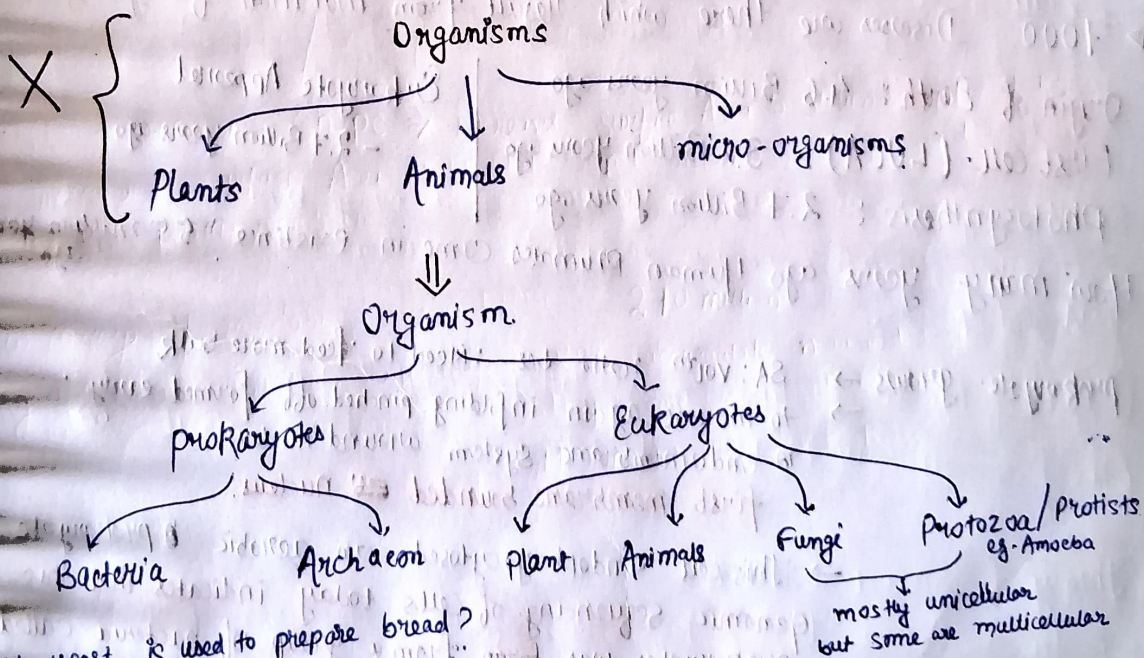
~~A network~~ A network of protein fibers that helps cells maintain their shape and internal organization. It is present in the cytoplasm. It is made up of three main types of filaments: actin, microtubules and intermediate filaments.

Prokaryotes have higher surface Area to Volume ratio.
→ Higher metabolic rate → shorter generation time

Eukaryotes

Eu → well defined

- i) presence of well-defined nucleus.
- ii) Presence of membrane-bounded organelles.
- iii) Presence of cyto-skeleton
- iv) Uni and multicellular.
- v) linear DNA.
- multiple chromosomes.
- vi) 10-100 μ m
- vii) 1.5×10^7 to 5×10^9
- viii) reproduce ~~but~~ through either of two.
- ix) absence of plasmid (except chromosomal material)
↳ Antibiotic resistance immunity against various pathogens



Which yeast is used to prepare bread?

Active dry yeast or Instant yeast

Saccharomyces Cerevisiae.

Common things between prokaryotes and Eukaryotes:
1. Presence of ~~cell~~ cell membrane, cytoplasm and ribosomes, genetic material.

How Antibiotic Work on bacteria?

(i) Reproduction Destruction

(ii) Protein production Sabotage

~~Not~~ (i) Cell wall Demolition

Note - plasmid has antibiotic resistance gene.

virus that kills bacteria are called bacteriophage.

Note - Cell wall maintain Homeostasis.

Endosymbiotic Theory

The endosymbiotic theory is a theory that explains how eukaryotic cells may have evolved. It states that eukaryotic cells evolved from simpler prokaryotic cells that merged together.

Who proposed the Endosymbiotic theory?

ans - Lynn Margulis

► 7000 Diseases are there which result from mutation of single gene.

Origin of Earth: 4.6 Billion years ago

First cell (LUCA): 3.8 Billion years ago

Photosynthesis: 2.7 Billion years ago

Eukaryote Appeared

- 2.7 Billion years ago

How many years ago Human Primates came in existence? 55 million years ago

prokaryote grows → SA: vol^m ratio dec. : Need to feed more bulk to increase again infolding pinched off formed early.
→ endomembrane system around nucleus.
→ first membrane bounded ~~cell~~ nucleus.

→ This cell endophagocytosed. i.e. aerobic prokaryote
Genomic sequencing of cells today indicates that this prokaryotes was from a group of bacteria called Alpha proteo bacteria

~ Mitochondria

⇒ Animal Cell

Some of these cells endo-phagocytose some Cyanobacteria

~ Chloroplast

→ Plant Cell

Mitochondria & Chloroplast ~ replicate on their own using process similar to binary fission

↓
Come from mother only

~ presence of 70S ribosomes

~ ~~some have~~ two membranes

~ Location of Electron Transport System

~ They have their own DNA + circular DNA

~ genome is similar to bacteria

~ Pores & Cardiolipin are only found in mitochondria chloroplast & Bacteria

~ Small size

E. coli - Human } Endosymbiosis : ~~we gave~~ we give them humid & shelter

~ They synthesize Vitamin K

~ also break down cellulose.

Termites ~ have prokaryotic bacteria in their gut that help them digest wood.

~ 1.8 billion years

Paramecium Bursaria ~ contain algae inside its cytoplasm. which provide it sugar & energy

~ other hand Par Bursaria provides protection for the algae eaters and viruses.

Jestate Amoeba with endosymbiotic algae.

Anaerobs ~ organism that don't require O_2 in fact O_2 can kill them.

↳ eg Metopus : endosymbiotic relationship with Methanogenic Archaea.

Hydra + Algae

69% energy.

How many years ago Human Primates came into existence?
Ans - 55 million years ago

99.97% Genome of primates & Human match.

Protein that converts DNA $\rightarrow$ RNA in our body?

LUCA - Last Universal Common Ancestor

- Evolution

process of gradual change that takes place over many generations, during which species of animals, plants, or insects slowly changes some of their physical physical characteristics.

Evolutionary process gives rise to diversity at every level of biological organization, including species, individual organism and molecules such as DNA and proteins.

Origin of life on Earth is also called Proto biogenesis.
The first cell called protocell came into existence about 3.8 billion years ago.

1. Theory of Special Creation

Earth 4000 years old } The diversity of life forms was always the same
- Creation by GOD. } Since the time of creation & will remain the same in future also.

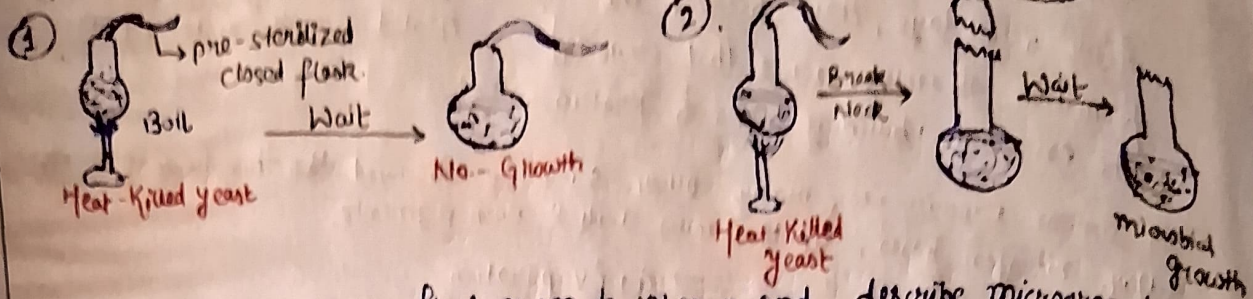
2. Theory of Spontaneous Generation

- Life arise from non-living matter.
- Aristotle

- dust creating flea

- maggots are arising from rotting meat

Louis Pasteur proposed "Life comes from life"



Anton van Leeuwenhoek ~ first person to observe and describe microorganism

Francesco Redi ~ challenged this theory.

3. Theory of ~~the~~ chemical Evolution.

British Scientist -

JBS Haldane

Russian Biochemist -

~~Pranav Borkar~~

Aleksandr Oparin

Book: The origin of life

Organic molecules could be formed from abiotic materials in the presence of an external energy source (eg. UV)

+ primitive atmosphere was reducing and contained ammonia and water vapour among other gases

Both also suspected that the first life-forms appeared in the warm, primitive ocean and were **Heterotrophic** (obtained preformed nutrients from the compounds in existence on early Earth). ~~rather~~ rather than autotrophic

They believed that simple organic molecules first formed and in the presence of UV become increasingly complex, ultimately forming cells.

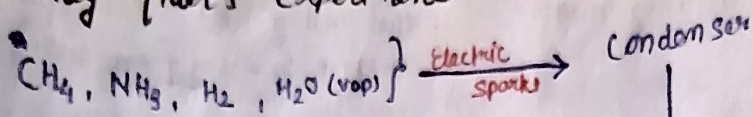
Strong Reducing conditions { inorganic molecules $\Rightarrow$ organic molecules (RNA, proteins, etc)
formation of life preceded by chemical evolution

Early Atmosphere - No O_2
 $\text{CO}_2 + \text{N}_2$
+ { $\text{H}_2, \text{H}_2\text{S}, \text{CO}$ }

} Reducing condition

+ Source of ~~extra~~ Energy such as sunlight or electrical discharge

Stanley Miller's Experiment.



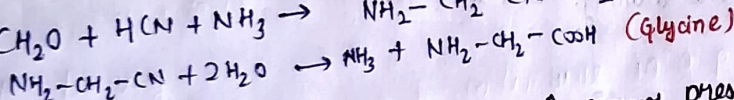
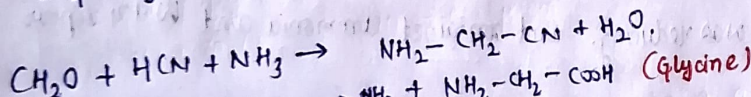
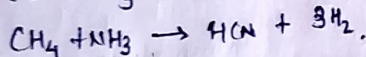
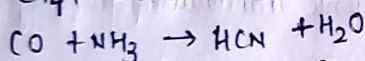
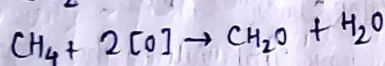
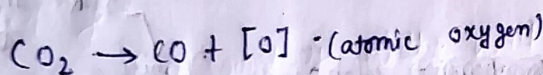
Chamber was heated provide water vapour

After a few days, they observed the formation of amino acids that serve as the essential building blocks of protein.

In similar experiments, many other scientists observed the formation of sugar, nitrogen bases, lipids, amino acids & even pigments.

Stages of Water Cycle

- (i) Evaporation
- (ii) Condensation
- (iii) Precipitation
- (iv) Deposition or Collection



Glycine, Glutamic Acid, Aspartic Acid, Alanine } present in sample.

Cosmozoic Theory:

Life come on the earth from outspace through meteoroids and/or dust grains and/or comets in the form microscopic life (spores and microorganisms).

This is called 'Cosmozoa' or 'Panbiogenesis' which were preserved inside the

random meteorites

which were stuck into barren earth to release cosmozoa $\Rightarrow$ present day life forms.

~~Some~~ ~~two~~ ~~beliefs~~

two beliefs

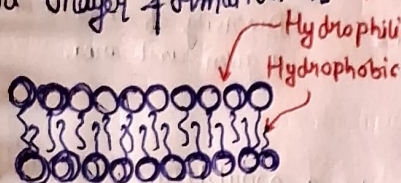
first biopolymers may have formed near

Hydrothermal Vents

Heat emanated from Hot ~~core~~ core of Earth

Catalyzing chemical Reaction

Lipid bilayer formation is **thermodynamically favourable** and $\therefore$ spontaneous



Evolution: Change over time.

Biological Evolution: changes in things which are able to reproduce

$\rightarrow$ may lead to development of new abilities in a species

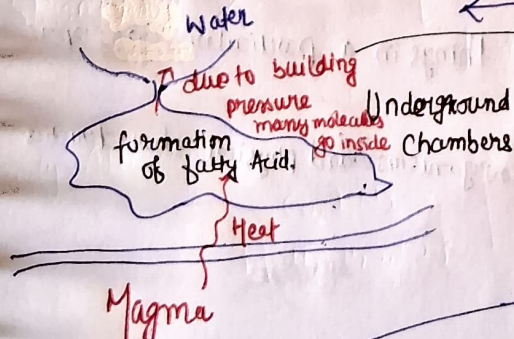
Three Conditions:

① Reproduction

② Variation

③ Selection

All kinds of variation of species $\rightarrow$ Only few survive which are fittest by natural conditions around (natural selection) $\sim$ **survival of fittest**



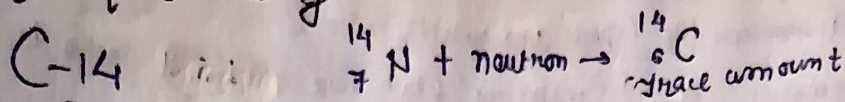
Natural Selection

Some molecules comes up
Some ~~settle~~ settle down

fatty Acid float in water

$\rightarrow$ They can come closer to form ball like structure $\Rightarrow$ stable membrane

Radio Active Dating

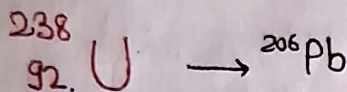


Dating.

Living organism ~ Same ratio of ^{12}C to ^{14}C as in Atmosphere

Dead organism ~ $^{14}_6\text{C} \xrightarrow{\text{Beta emission}} ^{14}_7\text{N}$ 5730 years old

Half life of ~~5730 years~~ $\rightarrow$



Half-life of 4.5 billion years

Biodiversity

Biodiversity ~ first coined by ~~Robert~~ Walter G. Rosen in 1985.

~ all of the population, species and communities in a defined area

- Types:
1. Genetic
Diversity of genes within a species
 2. Species
Diversity among species in an ecosystem
 3. Ecosystem.
Diversity of a habitat in a given unit area.

Genetic Diversity

Variation in genetic composition of individuals in a population, community or species.

High genetic diversity $\rightarrow$ increases the ability of the populations and species to survive in the major changes in their environment such as the climate change.

Species Diversity

Variety of species in an Ecosystem
Diversity of smaller organism is less known than the larger organism.

Ecosystem Diversity

Variation in all living and Non-living things in a particular Geographic or Ecological Region.

$\rightarrow$ unique combination of animals, plants, microorganisms and physical characteristics that define a location.

Why is Biodiversity Important?

- O_2 to Breathe
- Food to Eat
- Clothes to wear
- House to stay

Everything that exists in an ecosystem is part of web of life. Plants, animals, insects, interact and depend upon each other for what each offers, such as food, shelter, oxygen & soil enrichment.

Known Species: 1.7 to 1.8 million } 61% Insects
Total number of species: 5 to 30 million

Benefits: -

- > Food
- > Fuel
- > Medicines
- > Recreation
- > Crop Varieties
- > Industrial materials.
- > Education & Research
- > Traditional value

Non-consumptive value.

Threats :- Natural Causes:

- Narrow Geographical Area
- Over population
- Low breeding Rate
- Natural Disaster.

Anthropogenic (Human) Causes:

- Pollution
- Hunting
- Global warming & Climate change
- Agriculture

Biodiversity is measured in three terms:

α β γ

α $\rightarrow$ diversity within a particular Area or ecosystems
Usually expressed as species richness

β $\rightarrow$ comparison between ecosystems.

γ $\rightarrow$ overall diversity within a large Region

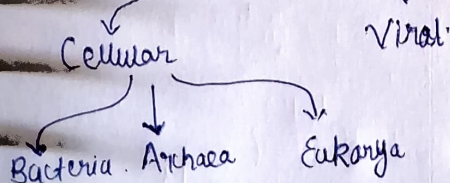
Genome

Human genome is diploid ~ two pairs of each chromosome one from each parent.

46 chromosomes ~ 23 pairs.

Human genome - 3.2 billion base pair of DNA } Reference
6.4 billion base pair.
 $\rightarrow$ only 1.5% 20,000 genes code for proteins.

Two Empire



Giant Viruses

~ are big enough to have their own parasites:

1. Human Immunodeficiency Virus (HIV)

~ 100 nm.

Genes: 9

Discovered: Human blood

2. Mimi Virus.

~ 400 nm

Genes: 1018

3. Moma Virus.

~ 400 nm

Genes: 1023

~ can be infected by
Sputnik virophage

~ 50 nm

Gene: 21

4. Megavirus

~ 440 nm

Gene: 1120

5. Pitho Virus.

~ 1500 nm.

Gene: 467

Discovered: 30,000 yrs old

Siberian ice core

E. coli Bacterium

~ 2000 nm

Gene: 4288

Discovered: Human colon.

1,40,000 Viruses present in our gut

No. of viruses > no. of species in the Earth.

Size of Brain is decided by viruses.

They encode some proteins essential for viral replication, but they never contain the full set of genes for proteins and RNAs required for translation, membrane function, or metabolism.

Viruses exploit cells to produce their components.

43% Human.

8% Viral DNA +

40% DNA are thought to have a viral origin

Reproduction

↳ Sexual

↳ Asexual.

~ integrate genetic material from both parents

Three domains of life

Evolutionary Lineage Tree

Eubacteria

Eukaryota

→ similar to

Archaea

Lipopolysaccharides

Outer membrane

Peptidoglycan layer

Periplasmic Space

Cytoplasmic membrane

porin

protein

Gram +ve

Gram -ve

Gram stain or Gram staining is a method of staining used to ~~class~~ classify bacterial species into two large groups: Gram-positive

Gram-negative.

Developed by Hans Christian Gram

Gram-Positive ~ thick mesh like cell wall made of peptidoglycan (50% - 90% cell envelope)

→ stained purple by crystal violet

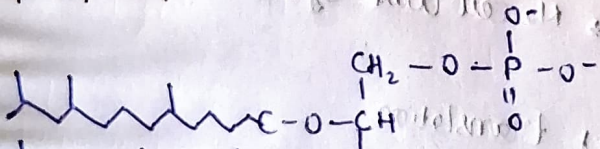
Gram-negative ~ thinner layer (10% cell envelope) so do not retain the purple stain and are counter-stained pink by safranin

Archaea v/s Eubacteria & Eukarya

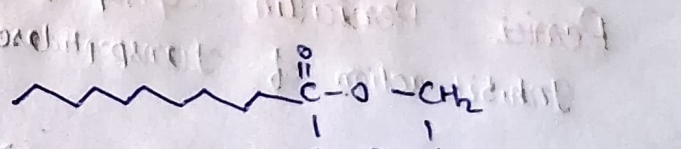
Archaea ~ ~~not~~ no peptidoglycan.

~ phospholipid

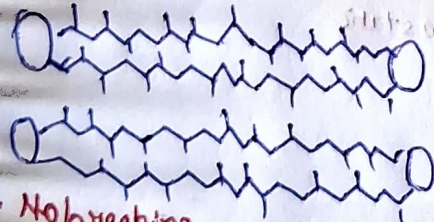
Eubacteria & Eukarya.



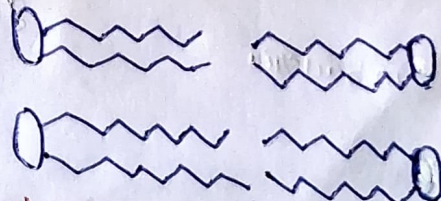
Ether linkage



Ester linkage



~ No breaking
~ ~~not~~ more branching



~ breaking
~ No branching

- Archaeal ~ main core membrane lipid is ~~ester~~ ether
- ~~Structure~~ One of the big features that distinguishes Archaea from bacteria and Eukarya is their membrane lipids were ~~not~~ archaeal play an important role.
- Archaeal is also broadly used as biomarker for ancient climates

Antibiotic

- ~ discovered by Alexander Fleming
- ~ Penicillin

Mechanism:

- (i) disrupt cell membrane integrity
- (ii) inhibit Nucleic Acid Synthesis
- (iii) inhibit protein synthesis
- (iv) inhibit cell wall synthesis

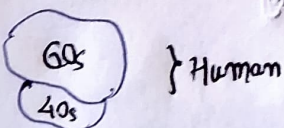
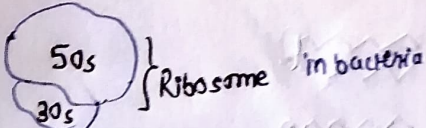
① Inhibit Cell Wall Synthesis:

~ peptidoglycan
have inter-layer peptide bond ~ without them the cell wall become unstable
~ cell can burst due to ~~High~~ High osmotic pressure.

~~Penicillin~~ Penicillin (Group of β -lactams) Bactericidal
Inhibit action of Transpeptidase $\rightarrow$ No cell wall synthesis

② Inhibit Protein Synthesis $\rightarrow$ Affect translation

Tetracycline (Tetracyclines) ~ interferes with 30s subunit
~ depletes bacteria from protein required for growth Bacteriostatic



③ Inhibit Nucleic Acid Synthesis

Rifamycin ~ inhibit activity of RNA polymerase

Antibiotic Resistance ~ Caused by mutations
~ may lead to multiple drug resistance.

Steps of Gram Staining:

1. Prepare thin smear of culture on a clean glass slide (oil free)
2. Heat fix by exposing the smear to indirect flame
3. Add few drops of crystal violet on the smear
~ wash the crystal with water after sometime
~ At this stage all cells appear in purple colour
4. Add plenty of Gram's Iodine to the smear
~ Rinse with water
~ Cells appear purple yet
5. ~~Decolourise~~ Decolourise with 95% Ethyl Alcohol for < 30 sec
~ wash with water
~ Gram +ve - purple
~ Gram -ve - colourless

Upper layer in Gram negative get washed away
6. Add Safranin
~ wash with water
~ Gram +ve - purple
~ Gram -ve - pink.

Disadvantage ~ Species identification is not possible

Note: ~ Organisms ~ Bacteria Eukarya } Swap gene with each other over a period of time
Archaea } ~ lateral Gene transfer

Archaea ~ Extremophiles

~ Carl Woese discovered
Wolfram Zillig ~ found Archaea are more related to Eukarya
by comparing 16S Ribosomal DNAs of Archaea & Bacteria.

Agon Asgard & Loki Archaea ~ have Eukaryotic Signature Proteins
which are only present in Eukarya

~ Bacteriophages produce endolysin which burst the bacteria when
enough bacteriophages are present in bacteria.

- Retro Viruses embed instructions for making new viral products in our DNA
- Sometime they get eliminated by WBCs (lymphatic system)
- But viruses that infect germ cells
 - ↳ DNA was passed to offspring then all Modern human had instruction for making certain types of viruses.

~ Human Endogenous Retroviruses

(HERVs) ~ 8% of DNA

HERV-H ~ make pluripotent for some time

HERV-K Helps embryos develop a built in Immune System that keep them safe even before they develop antibodies to pathogens in outside world.

HERV-E ~ converted one of the genes that codes for amylase in our pancreas into a salivary gland version